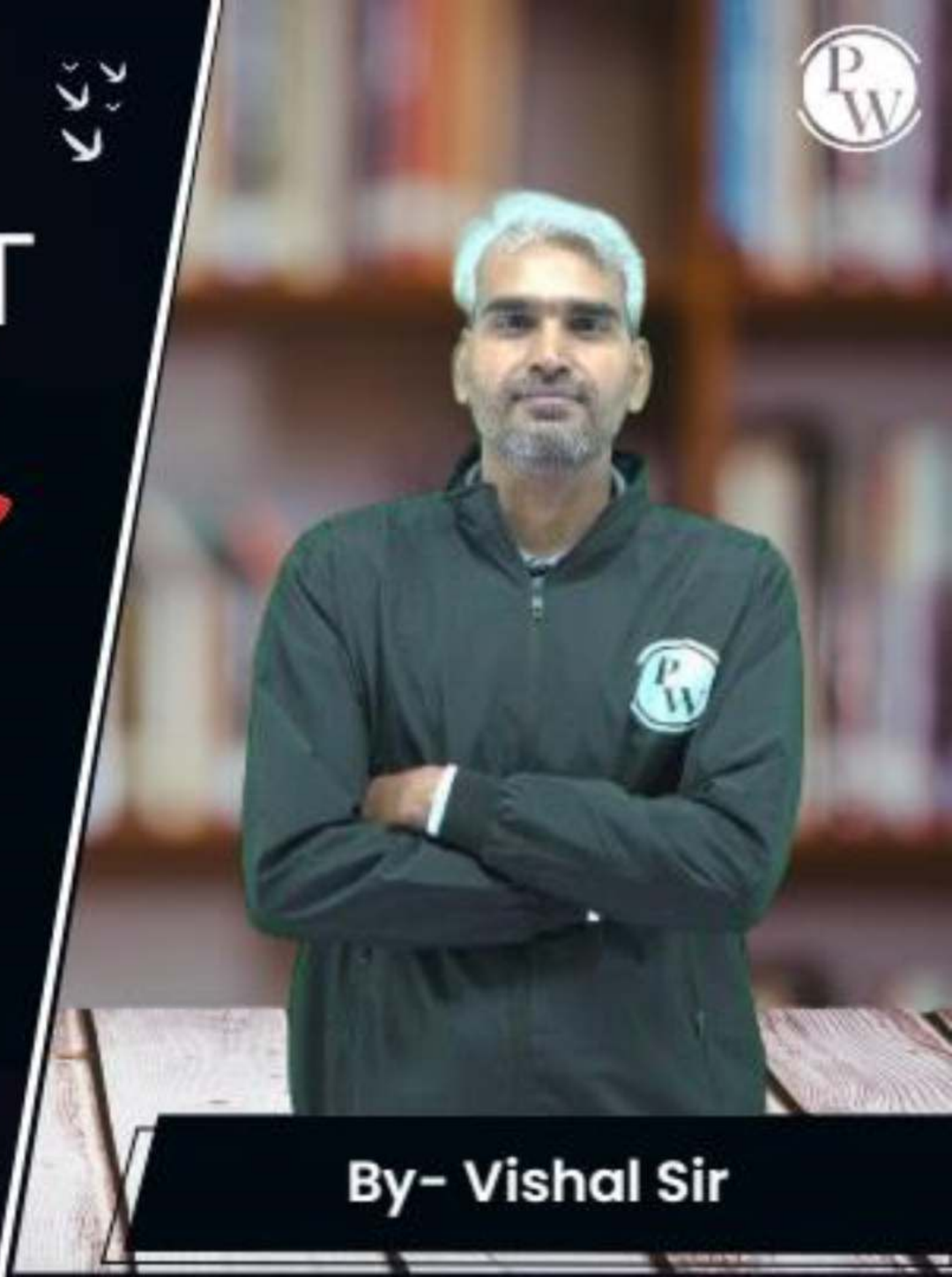
# Computer Science & IT



## Discrete Mathematics

## Set Theory & Algebra

**Lecture No. 21**



By- Vishal Sir

# Recap of Previous Lecture



Topic — Identity function and constant function

Topic — Identical function

Topic — Function composition

Slide

# Topics to be Covered



- Topic: Basic concepts associated with group
- Topic: Finite group
- Topic: Addition modulo ‘m’
- Topic: Multiplication modulo ‘m’

Slide



# Topic : Special Sets

$N$ = Set of all natural numbers

$Z$ = Set of all integers

$R$ = Set of all real numbers

$Q$ = Set of all rational numbers

$Q^*$ = Set of all non-zero rational numbers

Slide



# Topic : Algebraic Structure

## / Groupoid

Let 'S' be any non-empty set, and '*' is some binary operation,

then set S w.r.t. binary oph '*' is called a groupoid if and only if

$$a * b \in S, \quad \forall a, b \in S$$

↳ it is called Closure property.

i.e. Set S is closed w.r.t. binary oph '*'

* Set 'S' w.r.t. binary oph '*' can be denoted by $(S, *)$

Slide

| | Algebraic Structure (Groupoid) | Semi-Group | Monoid | Group | Abelian Group |
|---|---|---|---|---|---|
| $(\mathbb{N}, +)$ | ✓ | | | | |
| $(\mathbb{N}, \cdot)$ | ✓ | | | | |
| $(\mathbb{N}, -)$ | ✗ | | | | |
| $(\mathbb{N}, \div)$ | ✗ | | | | |
| $(\mathbb{Z}, +)$ | ✓ | | | | |
| $(\mathbb{Z}, \cdot)$ | ✓ | | | | |
| $(\mathbb{Z}, -)$ | ✓ | | | | |
| $(\mathbb{Z}, \div)$ | ✗ | | | | |
| $(\mathbb{Q}, +)$ | ✓ | | | | |
| $(\mathbb{Q}, \cdot)$ | ✓ | | | | |
| $(\mathbb{Q}, -)$ | ✓ | | | | |
| $(\mathbb{Q}, \div)$ | $0 \in \mathbb{Q}$ & $0 \div 0$ is not defined (∴ Not closed) | | | | |
| $(\mathbb{Q}^*, +)$ | $\left(\frac{p}{q}\right)+\left(\frac{-p}{q}\right)=0 \notin \mathbb{Q}^*$ ∴ Not closed | | | | |
| $(\mathbb{Q}^*, \cdot)$ | ✓ | | | | |
| $(\mathbb{Q}^*, -)$ | $\left(\frac{p}{q}\right)-\left(\frac{p}{q}\right)=0 \notin \mathbb{Q}^*$ ∴ Not closed | | | | |
| $(\mathbb{Q}^*, \div)$ | ✓ | | | | |

$\mathbb{N}$ → $(\mathbb{N}, +)$

$\mathbb{Z}$ → $(\mathbb{Z}, +)$

$\mathbb{Q}$ → $(\mathbb{Q}, +)$

$\mathbb{Q}^*$ → $(\mathbb{Q}^*, +)$

Slide



# Topic : Semi-group

A groupoid $(S, *)$ is called a semi-group if and only if following property holds true

$$(a*b)*c = a*(b*c) \quad , \forall a, b, c \in S$$

↳ i.e. operation must be associative

Slide

| | Algebraic Structure (Groupoid) | Semi-Group | Monoid | Group | Abelian Group |
|---|---|---|---|---|---|
| (N, +) | ✓ | ✓ | | | |
| (N, ·) | ✓ | ✓ | | | |
| (N, −) | ✗ | ✗ | | | |
| (N, ÷) | ✗ | ✗ | | | |
| (Z, +) | ✓ | ✓ | | | |
| (Z, ·) | ✓ | ✓ | | | |
| (Z, −) | ✓ | ✗ | | | |
| (Z, ÷) | ✗ | ✗ | | | |
| (Q, +) | ✓ | ✓ | | | |
| (Q, ·) | ✓ | ✓ | | | |
| (Q, −) | ✓ | ✗ | | | |
| (Q, ÷) | ✗ | ✗ | | | |
| (Q*, +) | ✗ | ✗ | | | |
| (Q*, ·) | ✓ | ✓ | | | |
| (Q*, −) | ✗ | ✗ | | | |
| (Q*, ÷) | ✓ | ✗ | | | |

N

Z

Q

Q*

Slide



# Topic : Monoid

A semi-group $(S, *)$ is called a Monoid if and only if there exists an element "$e \in S$" s.t

$$a * e = a, \forall a \in S$$

and

$$e * a = a, \forall a \in S$$

↳ Such an element '$e$' is called identity element w.r.t. binary op$^n$ '$*$'

Slide

| | Algebraic Structure (Groupoid) | Semi-Group | Monoid | Group | Abelian Group |
|---|---|---|---|---|---|
| (N, +) | ✓ | ✓ | $0 \notin N$ ∴ Not Monoid | (Identity w.r.t addition is '0') | |
| (N, ·) | ✓ | ✓ | ✓ $1 \in N$ | | |
| (N, −) | ✗ | ✗ | ✗ | | |
| (N, ÷) | ✗ | ✗ | ✗ | | |
| (Z, +) | ✓ | ✓ | ✓ | | |
| (Z, ·) | ✓ | ✓ | ✓ | | |
| (Z, −) | ✓ | ✗ | ✗ | | |
| (Z, ÷) | ✗ | ✗ | ✗ | | |
| (Q, +) | ✓ | ✓ | ✓ | | |
| (Q, ·) | ✓ | ✓ | ✓ | | |
| (Q, −) | ✓ | ✗ | ✗ | | |
| (Q, ÷) | ✗ | ✗ | ✗ | | |
| (Q*, +) | ✗ | ✗ | ✗ | | |
| (Q*, ·) | ✓ | ✓ | ✓ | | |
| (Q*, −) | ✗ | ✗ | ✗ | | |
| (Q*, ÷) | ✓ | ✗ | ✗ | | |

N

Z

Q

Q*

Slide



# Topic : Group

A monoid $(S, *)$ is called a group, if and only if, for every element $a \in S$ there exists some element $b \in S$, such that

$$a \underset{(\text{and})}{*} b = e \text{ (identity)}$$
$$b * a = e$$

↳ Here 'a' & 'b' are called inverse of each other

i.e. for a group, inverse of every element of the set must exist within the set itself!

Slide

| | Algebraic Structure (Groupoid) | Semi-Group | Monoid | Group | Abelian Group |
|---|---|---|---|---|---|
| (N, +) | ✓ | ✓ | ✗ | ✗ | |
| (N, ·) | ✓ | ✓ | ✓ $1 \in N$ | ✗ ← inverse does not exist for any Element, Except for '1' | |
| (N, −) | ✗ | ✗ | ✗ | ✗ | |
| (N, ÷) | ✗ | ✗ | ✗ | ✗ | |
| (Z, +) | ✓ | ✓ | ✓ | ✓ | |
| (Z, ·) | ✓ | ✓ | ✓ | ✗ ← Inverse does not exist for any other element. Except for '1' & '−1' | |
| (Z, −) | ✓ | ✗ | ✗ | ✗ | |
| (Z, ÷) | ✗ | ✗ | ✗ | ✗ | |
| (Q, +) | ✓ | ✓ | ✓ | ✓ | |
| (Q, ·) | ✓ | ✓ | ✓ | ✗ $0 \in Q$, inverse of '0' does not Exist w.r.t. multiplication | |
| (Q, −) | ✓ | ✗ | ✗ | ✗ | |
| (Q, ÷) | ✗ | ✗ | ✗ | ✗ { A set Containing '0' Can Never be a group w.r.t. multiplication } | |
| (Q*, +) | ✗ | ✗ | ✗ | ✗ | |
| (Q*, ·) | ✓ | ✓ | ✓ | ✓ | |
| (Q*, −) | ✗ | ✗ | ✗ | ✗ | |
| (Q*, ÷) | ✓ | ✗ | ✗ | ✗ | |

N → (N, +)

Z → (Z, +)

Q → (Q, +)

Q* → (Q*, +)

Slide



# Topic : Abelian group / Commutative group

A group $(S, *)$ is called an abelian group if and only if,

$$a * b = b * a \quad , \quad \forall a, b \in S$$

↳ i.e, op$^n$ '$*$' must be commutative on the elements of set S.

Slide

| | Algebraic Structure (Groupoid) | Semi-Group | Monoid | Group | Abelian Group |
|---|---|---|---|---|---|
| (N, +) | ✓ | ✓ | ✗ | ✗ | ✗ |
| (N, ·) | ✓ | ✓ | ✓ $1 \in N$ | ✗ | ✗ |
| (N, −) | ✗ | ✗ | ✗ | ✗ | ✗ |
| (N, ÷) | ✗ | ✗ | ✗ | ✗ | ✗ |
| (Z, +) | ✓ | ✓ | ✓ | ✓ | ✓ |
| (Z, ·) | ✓ | ✓ | ✓ | ✗ | ✗ |
| (Z, −) | ✓ | ✗ | ✗ | ✗ | ✗ |
| (Z, ÷) | ✗ | ✗ | ✗ | ✗ | ✗ |
| (Q, +) | ✓ | ✓ | ✓ | ✓ | ✓ |
| (Q, ·) | ✓ | ✓ | ✓ | ✗ | ✗ |
| (Q, −) | ✓ | ✗ | ✗ | ✗ | ✗ |
| (Q, ÷) | ✗ | ✗ | ✗ | ✗ | ✗ |
| (Q*, +) | ✗ | ✗ | ✗ | ✗ | ✗ |
| (Q*, ·) | ✓ | ✓ | ✓ | ✓ | ✓ |
| (Q*, −) | ✗ | ✗ | ✗ | ✗ | ✗ |
| (Q*, ÷) | ✓ | ✗ | ✗ | ✗ | ✗ |

N

Z

Q

Q*

Slide

## Note

1. Inverse of identity element will be identity element itself.
2. Identity element (if exists) in the set, then it will be unique w.r.t. given binary op$^{n}$ '$*$'.
3. In a group inverse of every element of the set exists within the set itself.
   and if $inv(a) = b$ then $inv(b) = a$ i.e. $(a^{-1})^{-1} = a$
4. In a group $(G, *)$, $(a * b)^{-1} = b^{-1} * a^{-1}$ always holds true $\forall a, b \in G$

## Groupoid

1. Closed

## Semi-group

1. Closed
2. Associative

## Monoid

1. Closed
2. Associative
3. Identity element must be present within the set

## Group

1. Closed
2. Associative
3. Identity element must be present within the set
4. Inverse of every element of the set must be present within the set

## Abelian group

1. Closed
2. Associative
3. Identity element must be present within the set
4. Inverse of every element of the set must be present within the set
5. Commutative

HoWo

#Q. Let $A = \{0, \pm 2, \pm 4, \pm 6, .....\}$

$B = \{0 \pm 1, \pm 3, \pm 5 .....\}$

Which of the following is not a semi- group

A $(A, +)$

B $(A, \bullet)$

C $(B, +)$

D $(B, \bullet)$

H.W.

#Q. Consider the set $\Sigma^*$ of all strings over the alphabet $\Sigma = \{0, 1\}$. $\Sigma^*$ with the concatenation operator for strings

A Not a semigroup

B Semi group but not a monoid

C Monoid but not a group.

D A group

HW

#Q. Let A be the set of all non-singular matrices of order n×n over real number and let * be the matrix multiplication operation. Then

A. A is closed under * but $\langle A, * \rangle$ is not a semigroup

B. $\langle A, * \rangle$ is a semigroup but not a monoid.

C. $\langle A, * \rangle$ is a monoid but not a group.

D. $\langle A, * \rangle$ is a group but not an abelian group.

H.W.

#Q. Let S be any finite set, and F(s) is defined as set of all function on set S. Then F(s) with respect to function composition operation (ie., o) is.

A Not a semigroup

B Semi group but not a monoid

C Monoid but not a group.

D A group

H.W.

#Q. Let Z is the set of all integers. The binary operation * is defined as a*b = max (a, b) then the structure (Z,*) is

A Not a semigroup

B Semi group but not a monoid

C Monoid but not a group.

D A group

H.W.

#Q. Let $Q^+$ be the set of all positive rational numbers. The binary operation * is defined as $a * b = \frac{ab}{3} \forall a, b, \in Q^+$ If $(Q^+, *)$ is a group then find

(i) identity element of the group

(ii) inverse of any element a, $\forall a \in$ Group

H.W.

#Q. Which of the following statement is/are not true.

A $\{0, \pm 2k, \pm 4k,, \pm 6k, ....\}$ is a group with respect to addition where k is any fixed positive integer

B $\{x \mid x \text{ is real number and } 0 < x \leq 1\}$ is a group with respect to multiplication

C $\{2^n \mid n \text{ is an integer}\}$ is a group with respect to multiplication

D None of these

# Finite Group

- A group $(G, *)$ is called a finite group if and only if set 'G' is a non-Empty finite set

- If $(G, *)$ is a finite group, then Order of group $(G, *)$ is same as Cardinality of set G!

i.e. $O(G) = |G|$

Order of group G

Note:

1. $\{0\}$ is a finite group of order = 1, w.r.t. addition
2. $\{1\}$ is a finite group of order = 1, w.r.t. multiplication.
3. $\{-1, 1\}$ is a finite group of order = 2, w.r.t. multiplication.

The above three groups are the only finite groups of real numbers w.r.t. addition and/or multiplication.

**Note:** If element 'e' is an identity element w.r.t. binary oph '*', then $\{e\}$ is always a finite group of 'order = 1' w.r.t. oph '*'.

**Note:** In a finite group of order = 2, Every element is inverse of itself.

**Note :-** Let 'S' is any non-Empty set.

Set 'S' is a group w.r.t. binary oph '$*$' if and only if.

{ ① $a * b^{-1} \in S, \ \forall a, b \in S$ ✓

and ② '$*$' is associative

Proof : Next Slide

This statement alone is sufficient to prove that

① 'S' is closed w.r.t. '$*$'

② Identity w.r.t. '$*$' is present in set S

③ Inverse of Every element of set S is present in set S itself.

* Let $a * b^{-1} \in S$, $\forall a, b \in S$.

## ① Identity :-

Let $a \in S$

$\therefore a, a \in S$

We know $\forall a, b \in S$

$a * b^{-1} \in S$

Hence if $a, a \in S$

then $a * a^{-1} \in S$

$\downarrow$

$e \in S$

i.e. identity element 'e' is present in set 'S'

## ② Inverse :-

① If identity element 'e' is the only element present in set S.

then we know $e^{-1} = e$

$\therefore$ inverse exist for every element of set S.

② If some other element $a \in S$, such that $a \neq e$

then $e, a \in S$

$\therefore e * (a^{-1}) \in S$

$\downarrow$

$a^{-1} \in S$

i.e. if $a \in S$ then $a^{-1} \in S$

## ③ Closed.

Let $a, b \in S$,

We just proved that

if $a, b \in S$ then $a^{-1}, b^{-1} \in S$

i.e if $a, b \in S$

then $a, b^{-1} \in S$

$\therefore a * (b^{-1}) \in S$

$\equiv$

$a * b \in S$

$\downarrow$

if $a, b \in S$

then, $a * b \in S$

Hence closed

eg: $1, \omega$ & $\omega^2$ are Cube roots of unity where $\boxed{\omega^3 = 1}$

* The set $\{1, \omega, \omega^2\}$ form a finite group of order = 3 w.r.t. binary opn multiplication.

"operation"

To obtain inverse we can use the concept of Composition table

**Closed** ① We know $\omega^3 = 1$, ∴ Set will be closed w.r.t. multiplication

**Associative** ② Multiplication is associative

**Identity** ③ Identity w.r.t. multiplication = 1 and $1 \in \{1, \omega, \omega^2\}$

**Inverse** ④ $\text{inv}(1) = 1$
$\text{inv}(\omega) = \omega^2$
$\text{inv}(\omega^2) = \omega$

| (Multiplication) | $1$ | $\omega$ | $\omega^2$ |
|---|---|---|---|
| $1$ | $1 = e$ | $\omega$ | $\omega^2$ |
| $\omega$ | $\omega$ | $\omega^2$ | $\omega^3 = 1 = e$ |
| $\omega^2$ | $\omega^2$ | $\omega^3 = 1 = e$ | $\omega^4 = \omega$ |

$\text{inv}(1) = 1$

$\text{inv}(\omega) = \omega^2$

$\text{inv}(\omega^2) = \omega$

eg:- $1, -1, i$ & $-i$ are four roots of unity.

We know $i^2 = -1$

The set $\{1, -1, i, -i\}$ form a finite group of order = 4, w.r.t. binary opn multiplication.

① $i^2 = -1$, ∴ Closed

② Multiplication is Associative

③ Identity w.r.t. multiplication = $1 \in \{1, -1, i, -i\}$

④ $\text{inv}(1) = 1$

$\text{inv}(-1) = -1$

$\text{inv}(i) = -i$

$\text{inv}(-i) = i$

* Note:- The set of 'n' roots of unity will form a finite group of order = n w.r.t. multiplication.



# Topic : Addition modulo 'm' $\oplus_m$

- Let 'm' is a fixed positive integer. (it gives remainder)
- For any two non-negative integers a & b

$$a \oplus_m b = \begin{cases} a+b & \text{if } (a+b) < m \\ r & \text{otherwise} \end{cases}$$

$\Rightarrow a \oplus_m b = (a+b) \% m$

$r$ is the remainder obtained on dividing $(a+b)$ by $m$

eg: $2 \oplus_6 3 = 5$

$4 \oplus_6 5 = 3$



# Topic : Multiplication modulo 'm' $\otimes_m$

- Let $m$ is a fixed Positive integer (it gives remainder)
- For any two non-negative integer $a$ & $b$

$$a \otimes_m b = \begin{cases} (a \cdot b) & \text{if } (a \cdot b) < m \\ \text{or} & \text{Otherwise} \\ (\text{remainder}) & \end{cases}$$

$$\equiv (a \cdot b) \% m$$

Eg: $2 \otimes_7 3 = 6$

$4 \otimes_7 5 = 6$



# 2 mins Summary

Topic Basic concepts associated with group

Topic Finite group

Topic Addition modulo ‘m’

Topic Multiplication modulo ‘m’

# THANK - YOU